SIMATS SCHOOL OF ENGINEERING
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Engineer to Excel

PLACEMENT & TRAINING CELL
Problem Statements-SET 3

1. Create a ABCDEIAN generator Using FOR LOOP

   Sample Output: Aate, Bate, Cate, Date, Eate, Fate, Gate , Hate.

2. Write a program to create a Caesar encryption and add a the ordinal number of the first character at the end of the string

   [Hint In cryptography, a Caesar cipher, also known as Caesar's cipher, the shift cipher, Caesar's code or Caesar shift, is one of the simplest and most widely known encryption techniques. It is a type of substitution cipher in which each letter in the plaintext is replaced by a letter some fixed number of positions down the alphabet. For example, with a left shift of 3, D would be replaced by A, E would become B, and so on. The method is named after Julius Caesar, who used it in his private correspondence]

   Sample Input: mee

   Sample Output: jbb107

   Test Cases:
   Unnam3, 2,004 Y#2000, 0.2221

3. Write a program to find 1st January is being a Sunday between 2014 and 2050

   Sample Output: 2023, 2034, 2040, 2045.

4. Write a program to calculate days left until Christmas and NEXT Christmas

   Sample Input

   Today's Date:15
   Current Month: November
   Year :2021

   Sample Output:
   This Christmas: 40 days
   Next Christmas:405 days

   Test Cases:
   12, AAA,0000
   10@, NOV, 20000
   13, December,0.0001
   25.0, December, 20000

5. Write a Python program to find four elements from a given array of integers whose sum is equal to a given number. The solution set must not contain duplicate quadruplets

   Sample Input and Output:
   Array Value:[-2,-1,1,2,3,4,5,6]

Target Value:10
[[-2,1,5,6],[-2,2,4,6][-2,3,4,5],[-1,1,4,6],[-1,2,3,6],[-1,2,4,5][1,2,3,4]]

6. Create a Pythagorean function
[Hint: The Pythagorean Theorem tells us that the relationship in every right triangle is: $c^2 = a^2 + b^2$, where c is the hypotenuse and a, b are two legs of the triangle.]

Sample Input
Enter A value:6
Enter B value:4
OUTPUT: It obeys the Pythagorean law.

Enter a Value 3
Enter B value 7

Output: It violates the Pythagorean law.

Test Cases: 2L, 20%, 1,000, 3.33

7. Write a function to create a UUID identifier.
(According to Wikipedia - A universally unique identifier (UUID) is an identifier standard used in software construction. A UUID is simply a 128-bit value. The meaning of each bit is defined by any of several variants. For human-readable display, many systems use a canonical format using hexadecimal text with inserted hyphen characters. For example: de305d54-75b4-431b-adb2-eb6b9e546014)

Sample Output
1fe35579-5ce7-46ec-89e0-7e723670029

8. Write a program to check if a number is a whole number or has a decimal place and also count the number of decimal digits.

Enter a Number: 9.000003

Output:
9.000003 is not whole number, It has 6 decimal places.

Test Cases:
00.00100#, 333,333.00, 2@year, 33.4USD

9. Write a program to convert a number in bytes to a human-readable string.

Enter the value in bytes:1000

Output:

1 KB

Test Cases:

2,000 0101010, 2.00002, 6782E

10. Write a Program to Calculate the hamming distance

[Hint:The Hamming distance between two strings, a and b is denoted as d(a,b).., Suppose there are two strings 1101 1001 and 1001 1101.11011001 ⊕ 10011101 = 01000100. Since, this contains two 1s, the Hamming distance, d(11011001, 10011101) = 2.]

Enter Number 1:10
Enter Number 2:7

Output :
Hamming distance is 3.

11. Write a program to check two given numbers and return true if one of the number is 50 or if their sum, product, difference, division is 50.

Sample Input:

Enter Number 1 20
Enter Number 2 30

Output:
TRUE